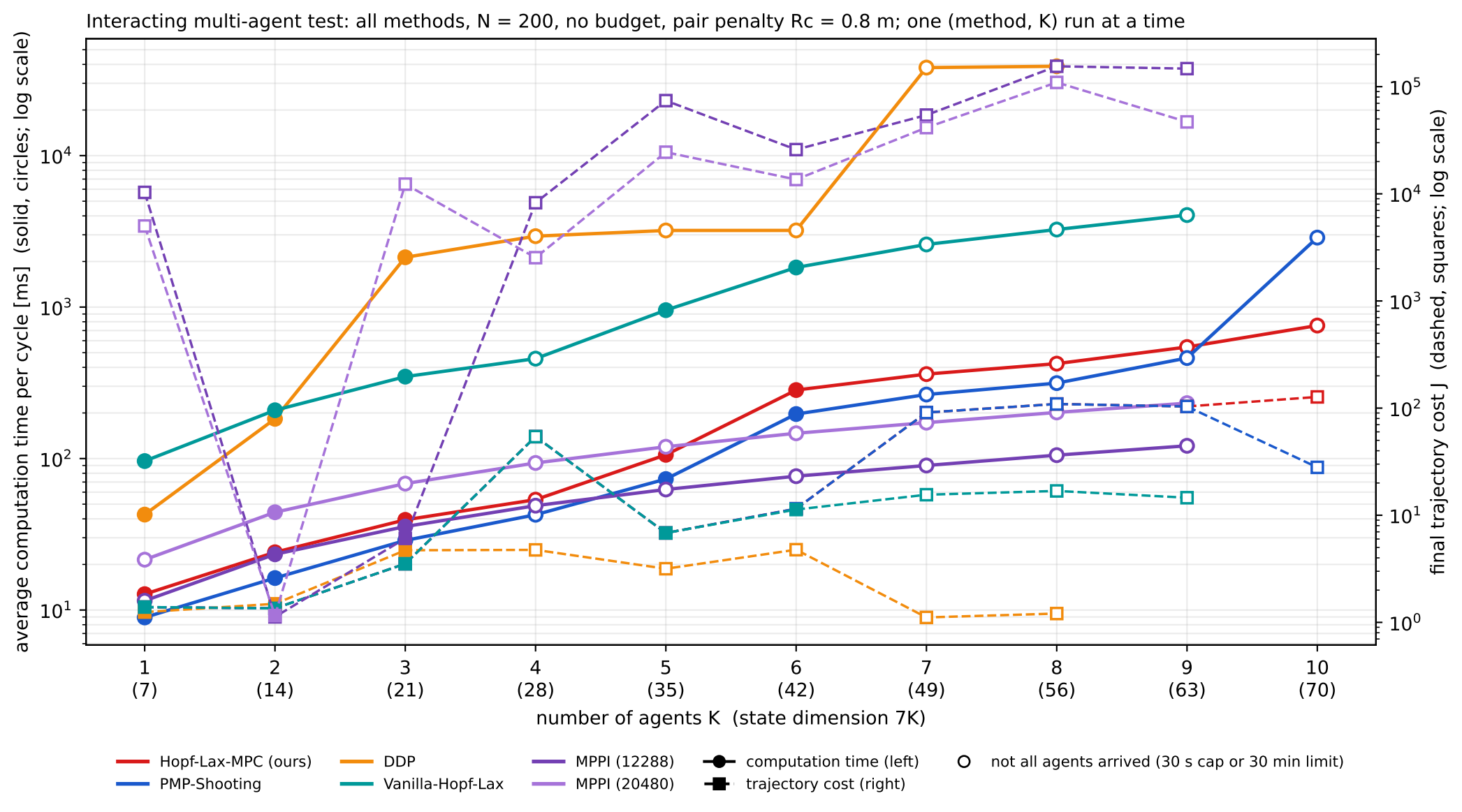
Interacting multi-agent test: all methods, N = 200, no budget, pair penalty Rc = 0.8 m; one (method, K) run at a time
average computation time per cycle [ms] (solid, circles; log scale)
final trajectory cost J (dashed, squares; log scale)
number of agents K (state dimension 7K)
1 (7)
2 (14)
3 (21)
4 (28)
5 (35)
6 (42)
7 (49)
8 (56)
9 (63)
10 (70)
Hopf-Lax-MPC (ours)
PMP-Shooting
DDP
Vanilla-Hopf-Lax
MPPI (12288)
MPPI (20480)
computation time (left)
trajectory cost (right)
not all agents arrived (30 s cap or 30 min limit)